

# 10.1 Missing data and data errors ↕

## □ Activity: Can you believe all that you read/see?

Read the following article on misleading data visualisations.

As you read, take time to consider and make notes on:

- how visualisations like these could be used to influence people
- what message(s) each visualisation conveys.

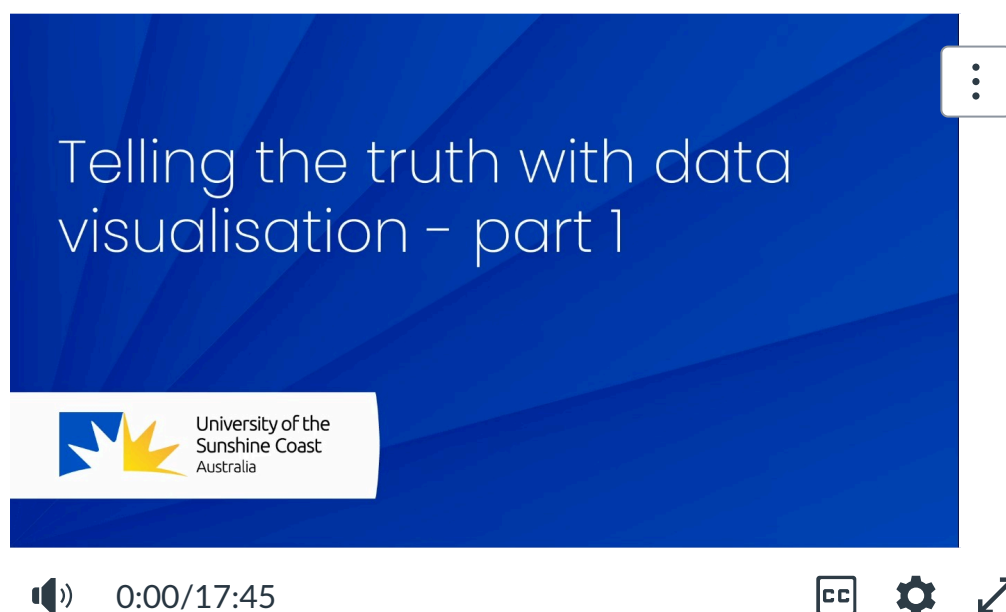
Explain your reasoning and bring your notes to the workshop.

Houbrechts, M. (2025, May 8). [Bad Data Visualization: 9 Examples to learn From](https://www.luzmo.com/blog/bad-data-visualization) [. \(https://www.luzmo.com/blog/bad-data-visualization\)](https://www.luzmo.com/blog/bad-data-visualization). Luzmo.

## □ Watch: Telling the truth with data visualisation (part 1) [17:45 mins]

Watch the video below, which explains how incorrect and/or incomplete data impacts the veracity of the data and therefore the value of a visualisation.

This is useful as it is important to understand that these types of mistakes can cause confusion and mislead audiences.



[Transcript](#)

<https://learn.usc.edu.au/courses/35242/files/31386>

[51?wrap=1](#)

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Download the following slides for note taking: [ICT702 Telling the truth with data visualisation \(part 1\)](https://learn.usc.edu.au/courses/35242/files/3168403?wrap=1) (<https://learn.usc.edu.au/courses/35242/files/3168403?wrap=1>)

Data visualisations are all created from data. Using poor quality data will result in misleading data visualisations and potentially inappropriate decisions. Many datasets contain missing data values and data errors. Knowing why these issues have occurred in a dataset and how to deal with them are important. Therefore, it is essential that the problems are identified and that the reason for their occurrence is understood. The way the data problems are dealt with impacts all subsequent analysis of the data.

## Identifying missing data

Missing data should be dealt with prior to dealing with data errors. In MS Excel, missing data can be identified various ways:

- using the *sort* function
- using the *conditional formatting* facility
- using the *COUNTBLANK* function.

How missing data instances are dealt with will depend on the variable involved and whether logical determination and/or validation of the missing value can be applied.

## Identifying data errors

Manual entry mistakes and miscalibration of sensors are two causes of data errors. Identifying outliers is one way of finding those data errors. Data errors can be identified in various ways:

- examining *summary statistics*
- Using MS Excel *MIN* or *MAX* functions to reveal unrealistic values
- using *frequency distributions*
- using *histograms*
- using *scatter charts*.

Not all data errors are extreme values, which makes them more difficult to identify. Scatter charts are a useful tool to utilise in these instances. It is important to be sure that an unusual value is erroneous, therefore using multiple methods of identification is often useful.